

Scalability & Future Plan

Date	29 September 2026
Team ID	SWTID-2026-2311
Project Name	AI StudyBuddy API
Maximum Marks	3 Marks

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The team's original ER diagram (see Project Documentation, ER Diagram Description) modeled Summary, Flashcard, Quiz and Study Plan as fully separate, normalized collections rather than the embedded fields actually shipped on the Material document this cycle. That original design doubles as the product roadmap:

Planned Change / Capability	Purpose	Status
Normalize Summary / Flashcard / Quiz / Study Plan into separate collections	Allow each AI resource to be queried, paginated and indexed independently of its source Material, and support multiple summaries/quizzes per material over time	Future roadmap (per original ER design)
Collaborative study groups	Let students share AI-generated resources with classmates, deferred from the original brainstorming session	Future roadmap
Additional upload formats (.docx, images with OCR)	Widen the range of study material students can upload beyond .txt/.md/.pdf	Future roadmap
Resource rating / feedback loop	Let students rate generated flashcards/quizzes so future Gemini prompts can be refined	Future roadmap
Async AI processing	Move the four Gemini-backed endpoints to a queued/async pattern so upload and generation don't block on Gemini's response time	Future roadmap